Kibum Kim

Ph.D Candidate

+82)10-2045-0438 | kb.kim@kaist.ac.kr | rlqja1107.github.io | kibum-kim-253b01206

Experience with Computer Vision

Computer Vision (Multimodal Learning)

- **(Current Work-*Video Understanding*)** Improving the inference efficiency of *Video MLLMs* for fine-grained understanding [21], streaming video understanding [25] and egocentric video understanding in edge-device applications (e.g., smart glasses).
- **(Recent Work-*Efficient Foundation Models*)** Token compression for MLLMs [21][22] and knowledge distillation for image MLLMs [19], CLIP-style video-language models [3], and their application (Recommender Systems [18]).
- **(Previous Works)** Focusing on scene understanding (i.e., the scene graph generation task), which converts images or videos into scene graphs to capture visual relationships between objects in the image or video domains.[5][7][9][10][12][13] .

Collaboration

- My research has been conducted in close collaboration with researchers across industry (Amazon [18], Oracle [21][2], ETRI [5][7][9][13][21], Huawei [25]) and academia (UC San Diego [18][21], Korea University [7][9][12][13], Seoul National University College of Medicine [ongoing]), resulting in co-authored publications at top-tier venues.

Research Vision

- My long-term vision is to build a real-time multimodal AI ecosystem for **wearable devices** (e.g., smart glasses) that understands the visual world precisely while responding instantly. Realizing on-device multimodal AI requires two capabilities in tandem, which define my research:

Fine-grained Visual Understanding

- On-device assistance demands a detailed understanding of a scene to perform tasks reliably. I first approached this through **scene understanding**, focusing on *long-tailed learning* to capture fine-grained visual relationships [5][7][9][10][12][13].
- More recently, I extend this to **multimodal LLMs**, improving fine-grained understanding via *compositional reasoning* [19][3] and *sink-token-aware token pruning* that preserves visual details [21].

Efficiency for Real-time Inference

- Wearable devices require low-latency responses under tight resource budgets. I therefore focus on **model compression** for multimodal LLMs, including *knowledge distillation* [19] and *token compression* [21][22][25].

Positions

Huawei R&D London — *London, UK*

Research Internship — *Mar 2026 - Sep 2026*

- Host: Principal Researcher: Zhensong Zhang
- Project: Video Multimodal LLMs for real-world applications (smart glasses)

University of California San Diego — *SD, USA*

Visiting Scholar in Computer Science Department — *Aug 2025 - Jan 2026*

- Host: Prof. Julian McAuley
- Project 1): [ICLR'26] Token-Efficient Item Representation via Images for LLM Recommender Systems [18]
- Project 2): [ECCV'26] Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs [21]

Education

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, South Korea*

Ph.D in Industrial & Systems Engineering — *Sep 2023 - Present*

- Research Interest: Efficient Multimodal Learning, Multimodal Large Language Model, Scene Understanding
- Advisor: Prof. Chanyoung Park

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
M.S IN INDUSTRIAL & SYSTEMS ENGINEERING *Aug 2021 - Jul 2023*

- Research Interest: Scene Understanding, Recommender systems, Graph Neural Network
- Advisor: Prof. Chanyoung Park

Hanyang University *Seoul, South Korea*
B.S. IN INDUSTRIAL ENGINEERING *Mar 2016 - Jul 2021*

- GPA: 4.09/4.5
- Early Graduation
- The period includes two years of military service, required for all Korean men

Publications

PREPRINT

[25] Kibum Kim, Dingli Liang, Yukai Huang, Haozheng Zhang, Weitong Cai, Rongguang Wang, Xiang Chen, Min Liu, Renjiu Hu, Gaolei Li, Yiqiao Xie, Jifei Song, Chanyoung Park, Zhensong Zhang, Hang Zhang. Structure-Preserving Visual Token Compression for Streaming Video Understanding via Bilateral Grid Splatting.

[24] (Under Review at AAAI'27) Kibum Kim, Kyle Min, Eduardo Escoto, Yueqi Wang, Julian McAuley, Chanyoung Park. When Data Scaling isn't Enough: Leveraging Video Scene Graphs to Improve Compositional Reasoning in Video-Language Models.

[23] (Under Review at WACV'27) SeonWoo Choi, Kibum Kim, Chanyoung Park, Donghyun Kim. Personalized Visual Representation Alignment for Generative Multimodal Recommendation. [Paper]

CONFERENCE

[22] (ECCV'26) Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park. Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding. [Paper]

[21] (ECCV'26) Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, Chanyoung Park. Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs. [Paper] [Code]

[20] (ICLR'26) Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, Chanyoung Park. IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra. [Paper]

[19] (ICLR'26) Jiwan Kim, Kibum Kim, Sangwoo Seo, Chanyoung Park. CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs. [Paper]

[18] (ICLR'26) Kibum Kim, Sein Kim, HongSeok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, Chanyoung Park. Token-Efficient Item Representation via Images for LLM Recommender Systems. [Paper] [Code]

[17] (NeurIPS'25) Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, Chanyoung Park. Training Robust Graph Neural Networks by Modeling Noise Dependencies. The Thirty-ninth Annual Conference on Neural Information Processing Systems. [Paper] [Code]

[16] (EMNLP'25-Findings) Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, Kibum Kim, Chanyoung Park. Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models. The 2025 Conference on Empirical Methods in Natural Language Processing. [Paper] [Code]

[15] (KDD'25) Sein Kim*, Hongseok Kang*, Kibum Kim, Jiwan Kim, Donghyun Kim, Minchul Yang, Kwangjin Oh, Julian McAuley, Chanyoung Park. Lost in Sequence: Do Large Language Models Understand Sequential Recommendation? ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. [Paper] [Code]

[14] (SIGIR'25) Jiwan Kim, Hongseok Kang, Sein Kim, Kibum Kim, Chanyoung Park. Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios. The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval. [Paper] [Code]

[13] (**ICLR 2025**) Kibum Kim, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon , Jinyoung Moon, Donghyun Kim, Chanyoung Park. Weakly supervised Video Scene Graph Generation with Natural Language Supervision. [Paper]

[12] (**AAAI 2025**) Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, Chanyoung Park. ReTAG: Retrieval-Augmented Scene Graph Generation via Multi-Prototype Learning. The 39th Association for the Advancement of Artificial Intelligence. [Paper] [Code]

[11] (**CIKM 2024**) Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, Chanyoung Park. Debiased Graph Poisoning Attack via Contrastive Surrogate Objective. ACM International Conference on Information and Knowledge Management.[Paper] [Code]

[10] (**ECCV 2024**) Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, Chanyoung Park. Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation. The 18th European Conference on Computer Vision ECCV 2024. [Paper] [Code]

[9] (**CVPR 2024**) Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park. LLM4SGG: Large Language Model for Weakly Supervised Scene Graph Generation. [Paper] [Code]

[8] (**WWW 2024 (Oral)**) Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, Chanyoung Park. Self-guided Robust Graph Structure Refinement. The 2024 ACM Web Conference. [Paper] [Code]

[7] (**ICLR 2024**) Kibum Kim*, Kanghoon Yoon*, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park. Adaptive Self-training Framework for Fine-grained Scene Graph Generation. The Twelfth International Conference on Learning Representations. [Paper] [Code]

[6] (**SIGIR 2023**) Kibum Kim, Dongmin Hyun, Sukwon Yun, Chanyoung Park. MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation. The 46th International ACM SIGIR Conference on Research and Development in Information Retrieval. [Paper] [Code]

[5] (**AAAI 2023**) Kanghoon Yoon*, Kibum Kim*, Jinyoung Moon, Chanyoung Park. Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network. Proceedings of the AAAI Conference on Artificial Intelligence 2023. [Paper] [Code]

[4] (**CIKM 2022**) Sukwon Yun, Kibum Kim, Kanghoon Yoon, Chanyoung Park. LTE4G: Long-Tail Experts for Graph Neural Networks. Proceedings of the 31st ACM International Conference on Information & Knowledge Management. [Paper] [Code]

Workshop

[3] (ECCV'26) Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, Chanyoung Park. Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs. Wearable AI (Best Paper Award) [Paper]

[2] (**NeurIPS'25)**) Kibum Kim, Kyle Min, Chanyoung Park. Data Scaling isn't Enough: Towards Improving Compositional Reasoning in Video-Language Models. Multimodal Algorithmic Reasoning (Spotlight) [LINK] and Efficient Reasoning.

[1] (**WWW'24 (Oral)**) Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, Chanyoung Park. Noise Robust Graph Learning under Feature-Dependent Graph-Noise. The Web Conference, Data-Centric Artificial Intelligence (DCAI) Workshop.

Lead Projects

2025.12–Present	**Foundation Model for Electromyography (EMG) signals in the Medical Domain** Collaboration with Seoul National University College of Medicine (Prof. Woo-Hyung Lee)
2021.06-Present	**Visual Intelligence Technique Development** Collaboration with **E**lectronics and **T**elecommunications **R**esearch **I**nstitute (ETRI)
2022.06-2026.05	**AI Development for reasoning, extraction, understanding of common-sense** Collaboration with **I**nstitute for **I**nformation & communications **T**echnology **P**lanning & evaluation (IITP)
2024.06-2025.07	**NAVER·Intel·KAIST (NIK) AI Collaboration Project for building a new AI ecosystem** Collaboration with NAVER & Intel
2020.12-2021.06	**Recommending Financial Product based on Graph Embeddings** Collaboration with Hana Bank

Professional Services

Conference Review

- 2026 - European Conference on Computer Vision (ECCV)
- 2026 - International Conference on Machine Learning (ICML)
- 2026 - The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- 2026 - The WebConf (WWW)
- 2025 - Neural Information Processing Systems Datasets and Benchmark Track (NeurIPS)
- 2025 ∼ 2026 - ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)
- 2025 ∼ 2026 - The International Conference on Learning Representations (ICLR)
- 2024 ∼ 2027 - AAAI Conference on Artificial Intelligence (AAAI)
- 2024 ∼ 2026 - Conference on Information and Knowledge Management (CIKM)

Workshop Review

- Efficient Large Vision Models [CVPR'25], Graph Learning with Foundation Models [PAKDD'25], Towards Robots with Human-Level Abilities [ICLR'25], Multimodal Algorithmic Reasoning [NeurIPS'25], Efficient Reasoning [NeurIPS'25]

Journal Review

- Neurocomputing, Journal of Big Data, Pattern Recognition Letters, TKDE, TNNLS, Knowledge-Based Systems

Awards & Scholarship

2026 **ECCV Workshop Best Paper Award**
Selected as a Best Paper Award at ECCV Wearable AI Workshop 2026

2026 **ICML Top Reviewer**
Selected as a Gold Reviewer at ICML 2026

2022 **Poster Competition Excellence Award**
Awarded at Industrial/Social Problem Solving Session held by Department of ISysE, KAIST

2022 **Hanyang Academic Achievement Award**
Awarded within the top 3% among the College of Engineering, Hanyang Univ.

2020 **Hanyang Brain Scholarship**
Scholarship for excellent top 5% grade in Industrial Engineering department, Hanyang Univ.

2017 **Hanyang Brain Scholarship**
Scholarship for excellent top 5% grade in Industrial Engineering department, Hanyang Univ.

References

Prof. Chanyoung Park, Assistant professor, KAIST
Email: cy.park@kaist.ac.kr

Prof. Julian McAuley, Professor, UC San Diego
Email: jmcauley@ucsd.edu

Kyle Min, Principal Applied Scientist, Oracle
Email: kyle.min@oracle.com